

Phase 3 – Project Design Phase

Project Title

Script-Controlled ACL – Restrict Record Access Based on Field Value

1. Introduction

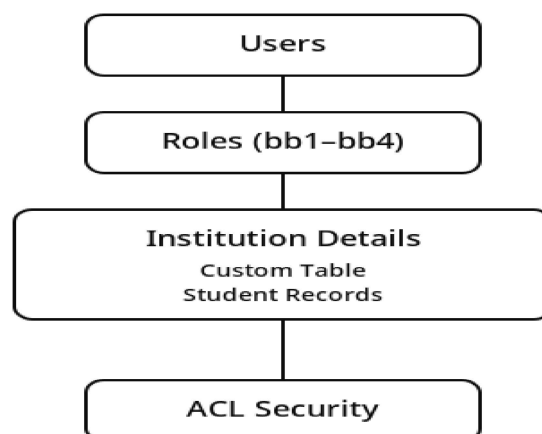
The Project Design phase defines how the ServiceNow application is structured before implementation. It explains the table design, field structure, user roles, ACL workflow, and system architecture used to secure the Institution Details records.

2. System Architecture

The application consists of four main components:

- Users (Admin and EEE User)
- Roles (bb1, bb2, bb3, bb4)
- Institution Details custom table
- Access Control Lists (ACLs)

Architecture Diagram



3. Table Design

Custom Table

Property	Value
Label	Institution Details
Name	u_institution_details
Extends	False

4. Field Design

Field Name	Data Type	Purpose
Student Roll Number	Auto Number	Unique ID
Student Name	Reference (User)	Student details
Faculty Name	Reference (User)	Faculty details
Branch	Choice	ECE, EEE, CSE
Email	String	Contact
Phone Number	String	Contact
Description	Multi-line String	Additional details

5. User and Role Design

User	Role	Permission
Admin	admin	Full access
EEE User	bb1	Read EEE records
EEE User	bb2	Create records
EEE User	bb3	Edit records
EEE User	bb4	Delete records

6. ACL Design

Read ACL

- Type: Record
- Operation: Read
- Table: u_institution_details
- Role: bb1
- Data Condition: Branch = EEE

Create ACL

- Role: bb2

Write ACL

- Role: bb3

Delete ACL

- Role: bb4

7. Workflow Diagram



8. Expected Design Outcome

The designed system ensures that access is controlled through user roles and the Branch field. The ACL structure provides secure record-level protection while maintaining full administrative control over the Institution Details application.